

Swagotam Malakar

Computer Science & Engineering Student

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Education

- 2022–2026 **B.Sc. in Computer Science & Engineering**, United International University, Dhaka, Bangladesh. (Expected)
- 2020 **Higher Secondary Certificate (HSC)**, Govt. Science College, Dhaka, Bangladesh, GPA: 5.00/5.00.
- 2018 **Secondary School Certificate (SSC)**, Jury Govt. Model High School, Sylhet, Bangladesh, GPA: 5.00/5.00.

Undergraduate Thesis (FYDP)

- Nov 2025–Present **Multi-Dimensional Center of Mass Estimation**, for Uniform and Non-Uniform Objects. Supervised Research (Thesis)
- Goal:** To develop a non-contact method for estimating the Center of Mass (CoM) of irregular, non-uniform objects using Computer Vision and Machine Learning, where classical physical balancing methods fail.
- Technical Achievements:**
- **3D Reconstruction:** Developing a pipeline to generate 3D mesh representations of physical objects from 2D camera feeds using photogrammetry techniques.
 - **Physics-Informed ML:** Implementing **PointNet++** for semantic segmentation of 3D point clouds to identify density variations in non-uniform objects.
 - **Data Pipeline:** Designing a robust sensor data processing workflow to handle high-dimensional spatial data for real-time estimation.

Independent Research

- Feb 2026–Present **FL Optimization for Edge-Assisted IoT Intrusion Detection**. Research Project (Final Phase)
- Goal:** To develop a robust, privacy-preserving intrusion detection system optimized for resource-constrained IoT edge devices.
- Technical Achievements:**
- **Algorithmic Innovation:** Formulated **MalakarFed-Prox**, a novel federated aggregation algorithm utilizing Z-Score normalized attention with temperature scaling to robustly penalize client drift and ensure global model convergence in highly heterogeneous (Non-IID) networks.
 - **Deep Learning Architecture:** Architected **DynamicEdgeMLP**, a lightweight, dimension-agnostic 1D neural network replacing conventional 2D CNNs, utilizing LayerNorm to prevent training crashes on edge nodes.
 - **Big Data Engineering:** Processed 8.6M+ network flows (TII-SSRC-23 dataset) via a highly memory-efficient data pipeline. Implemented dtype downcasting and chunked stratified sampling, reducing RAM consumption by ~50% while preserving minority class distributions.
 - **Simulation & Results:** Orchestrated a multi-client simulation using PyTorch, Ray, and Flower on distributed GPUs. Achieved a SOTA 98.66% overall accuracy and a 98.40% weighted F1-score under extreme class imbalance.

- Nov 2025–Present **Privacy-Preserving IDC Detection**, *via Federated Learning with GAN Augmentation*.
Ongoing Research Project
- Goal:** To develop a decentralized framework for detecting breast cancer (IDC) patches from histopathology images without centralized data sharing.
- Technical Achievements:**
- **Non-IID Handling:** Implemented **FedProx** and FedAvg algorithms to handle statistical heterogeneity across simulated hospital clients.
 - **GAN Augmentation:** Integrated **DCGAN** to generate synthetic minority class samples locally, resolving severe class imbalance (IDC-negative vs. positive).
 - **Model Optimization:** Utilized **EfficientNet-B0** with Focal Loss to maximize feature extraction efficiency on edge constraints.
 - **Results:** Achieved ~ 0.86 AUC with improved calibration ($ECE \approx 0.37$) on the global test set.
- Dec 2025–Present **Bornomala**, *Bangla Dialect Normalization Pipeline*.
Research Project
- Goal:** To bridge the performance gap of ASR systems on regional dialects (Sylheti, Chittagonian) by solving the "Language Identity Crisis" through domain-specific data acquisition.
- Technical Achievements:**
- **Data Mining Pipeline:** Developed an automated scraper to harvest unstructured dialect audio from ****YouTube**** (dramas/vlogs), augmenting the failure-prone RegSpeech12 dataset.
 - **Smart Segmentation:** Integrated ****Silero VAD**** to intelligently remove silence and background noise, creating high-fidelity speech segments for training.
 - **Transcription:** Deployed ****Whisper (Medium)**** for robust automated draft transcription, utilizing aggressive memory management for optimization on P100 GPUs.
- Aug 2025–Present **Swarabyanjan**, *Yellow Journalism Detection via Stance Detection*.
Research Project (Draft Almost Completed)
- Goal:** To automatically detect sensationalism and "clickbait" in Bengali news by verifying the semantic consistency between headlines and article bodies.
- Key Findings & Achievements:**
- **Benchmark Creation:** Curated and validated the "Bangla Headline-Body Consistency Benchmark (BHBC-1k)", a rigorous gold-standard dataset of ~ 800 mid-ambiguity news articles via multi-LLM consensus and expert human annotation from a 10,000+ corpus.
 - **LLM Ensemble & Bias Detection:** Engineered a validation pipeline aggregating zero-shot judgments from 23 LLMs (Meta, Google, Mistral, OpenAI, Alibaba). Designed a "smart bias-detection" algorithm penalizing skewed models, resolving 97.3% of cases automatically.
 - **Statistical Analysis:** Proved via McNemar's test ($\chi^2 = 106.54, p < 10^{-23}$) that cosine thresholding systematically over-flags legitimate news, demonstrating 93.9% precision on the majority class but a catastrophic 2.9% precision on the minority class.
 - **IAA Modeling:** Evaluated LLM-as-a-judge reliability against humans (Cohen's Kappa $+0.556$) supported by 2000-sample Bootstrap 95% CIs, establishing empirical evidence for LLM-in-the-loop label denoising.
 - **Tools utilized:** Python, Pandas, Scikit-learn, Matplotlib, Hypothesis Testing, LLM API Integration.

Major Projects

- Mar 2026–Present **ArduMedics – Smart Healthcare Assistant Robot**, *Machine Learning Developer*.
Overview: An autonomous healthcare robot integrating IoT, robotics, and AI for 24/7 contactless patient care and medical logistics.
Key Contributions (Machine Learning):
 - **AI-Powered Monitoring:** Developing a computer vision pipeline for real-time camera-based patient fall detection and automated emergency notifications.
 - **Prescription OCR:** Implementing Optical Character Recognition (OCR) models to extract medicine names, dosages, and schedules from handwritten prescription images.
 - **System Integration:** Integrating ML diagnosis and monitoring models with the Raspberry Pi 5 main controller and ESP32 medical hub for synchronized health data processing.**Tech Stack:** Python, Computer Vision (OCR), Deep Learning, Raspberry Pi 5, ESP32.
- Nov 2025–Feb 2026 **Cyber Aware Prohory**, *Machine Learning Developer*.
Overview: An automated classification system built to identify and filter text-based threats.
Key Features & Awards:
 - **Award:** Won the **Champion Prize** at the CSE Project Show (Fall 2025) in the Advanced Object-Oriented Programming (AOOP) category.
 - **Threat Detection Pipeline:** Developed and trained models to accurately categorize text inputs as spam, danger, or safe messages.
 - **Model Optimization:** Conducted iterative testing and evaluation of progressive model architectures (including versions v26, v27, and v28) across diverse text inputs to maximize classification performance.
- Jun 2025–Present **AmarShohor – Smart City Platform**, *Team Lead*.
Overview: A platform integrating civic engagement, real-time services, and utility connectivity.
Project Status: Currently in the high-fidelity prototyping phase. Complete design and user flows are finished in **Figma**, with full-stack development scheduled to commence soon.
Planned Features:
 - **Civic Services:** Issue reporting system, blood donation matching, and NGO/Utility connectivity.
 - **Smart Routing:** Implemented SOS alerts and dynamic map formulation using ML-powered priority management.**Tech Stack (Planned):** React Native, Node.js, Express.js, Django (ML), MongoDB, PostgreSQL.
- Oct 2024–Jan 2025 **Blockchain Supply Chain Integrity**, *Research & Concept Strategy*.
Overview: Researched and developed a framework for implementing blockchain technology to prevent counterfeit drugs in Bangladesh. (Project Video)
Key Contributions:
 - **Research & Analysis:** Authored a whitepaper assessing the pharmaceutical industry's vulnerabilities and the use of blockchain for end-to-end traceability.
 - **Strategic Framework:** Formulated strategies to overcome regulatory gaps and high initial cost barriers, aiming to automate compliance and mitigate organized crime risks.**Impact & Skills:** Designed to improve patient safety and regulatory adherence. *Skills: Research, Blockchain Architecture, Strategic Planning.*

Technical Skills

- Languages Python, C/C++, Java, JavaScript (ES6+), PHP, Solidity, LaTeX
- AI & Research Federated Learning (FedProx, Flwr), BanglaBERT, Whisper, Silero VAD, BM25, GANs, PyTorch, Ray
- Web & Tools React.js, Node.js, Django, Docker, AWS, Git, CI/CD, Linux

Certifications

Development	Stanford CS106A (Code in Place), HackerRank (Software Engineer), Cisco CLA (C Programming), HackerRank (Python Basic), Udemy (Java Programming), Udemy (LeetCode in Java)
Networking	CCNA: Introduction to Networks, Cybersecurity Essentials (Cisco), NDG Linux Essentials

Honours & Awards

Academic	Board Scholarship (PSC, JSC, SSC), Perfect GPA (5.00) in all public exams
Competitions	Multiple awards in National Science Olympiads and Inter-College Quiz Championships

Extracurricular Activities

Communities	UIU Competitive Programming Community - Active Member (2025-Present)
Social Service	Bangladesh Scouts - Active member (Jan 2010 - May 2018). Participated in community service projects, disaster relief operations, and environmental conservation.
Creative	Bengali Novelist - Authoring Bengali fiction exploring social philosophy and human identity.

Language Proficiency

English	Professional Working Proficiency	<i>Written: Excellent / Spoken: Fluent / Reading: Excellent</i>
Bangla	Native Speaker	<i>Mother tongue</i>
Hindi	Conversational	<i>Written: Poor / Spoken: Good / Reading: Basic</i>

Interests

Tech	AGI, Machine Learning (ML), Deep Learning (DL), Computer Vision, Blockchain, Cybersecurity
Research	Natural Language Processing, Bengali Language Computing, Data Mining, AI Ethics
Others	Astrophysics, Reading novels, Philosophy of science, History

References

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